

RANDOM GROUP DOCUMENTATION

ForensIQ

AI-Powered Transaction Intelligence for Financial Crime Detection

DSC Finance & Digital Innovation Hackathon 2026
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Project Report

1. Project Overview

ForensIQ is a data-driven fraud detection and transaction intelligence system developed for the DSC Finance & Digital Innovation Hackathon 2026.

The project analyses mobile money transaction data to identify suspicious financial activity, understand the behavioural patterns associated with fraud, and provide explanations that can support financial investigators.

The system is designed to move beyond simply asking:

“Is this transaction fraudulent?”

Instead, ForensIQ aims to answer:

“Why is this transaction suspicious, how risky is it, and what should an investigator do next?”

The project focuses on combining:

- Exploratory data analysis
- Feature engineering
- Machine learning
- Fraud pattern analysis
- Explainable risk indicators
- Business-oriented recommendations

The ultimate goal is to support financial institutions with faster, clearer, and more explainable transaction monitoring.

2. Problem Statement

Financial institutions process millions of transactions every day. Within this large volume of legitimate activity, a small number of transactions may be fraudulent.

This creates a major challenge:

Fraudulent transactions are rare, but the consequences of missing them can be significant.

A fraud detection system must therefore identify suspicious transactions while avoiding excessive false alarms.

The challenge is especially difficult because:

- Fraud represents only a very small percentage of transactions.
- Fraudulent behaviour may be hidden among legitimate activity.
- A transaction can appear normal when viewed in isolation.
- Traditional rule-based systems may fail to detect new or evolving patterns.
- Financial investigators need to understand why a transaction was flagged.

The ForensIQ project addresses this challenge by combining statistical analysis, engineered behavioural features, and machine learning to identify and explain suspicious transaction activity.

3. Dataset Description

The dataset consists of synthetic mobile money transactions modelled on real African financial services data.

The hackathon dataset is a stratified sample of approximately 630,000 transactions from a larger dataset of approximately 6.3 million rows. The dataset preserves a highly imbalanced fraud rate of approximately 0.13%.

The dataset contains transaction-level information including:

- Transaction type
- Transaction amount
- Origin account balance before the transaction
- Origin account balance after the transaction
- Destination account balance before the transaction
- Destination account balance after the transaction
- Fraud label
- Existing transaction flag

The original dataset contains five transaction types:

- PAYMENT
- TRANSFER
- CASH_OUT
- CASH_IN
- DEBIT

The analysis found that fraud is concentrated in:

- TRANSFER
- CASH_OUT

The other transaction types had no observed fraudulent transactions in the analysed dataset.

This does not mean that every TRANSFER or CASH_OUT transaction is fraudulent. Instead, these transaction types should be treated as higher-risk categories requiring additional behavioural analysis.

4. Understanding the Fraud Problem

Fraud detection is an extremely imbalanced classification problem.

The vast majority of transactions are legitimate, while only a very small proportion are fraudulent.

This creates an important modelling problem.

For example, a model that predicts every transaction as legitimate could achieve very high accuracy while detecting no fraud at all.

Therefore, accuracy alone is not a suitable evaluation metric.

The project focuses on metrics such as:

- Precision
- Recall
- F1-score
- PR-AUC
- ROC-AUC
- Confusion matrix
- False positives
- False negatives

The most important business question is not simply:

“How accurate is the model?”

Instead:

“How many fraudulent transactions can we detect, and how many legitimate customers will we incorrectly flag?”

5. Exploratory Data Analysis

The exploratory data analysis was used to understand how fraudulent transactions differ from legitimate transactions.

The analysis focused on:

- Fraud distribution
- Transaction type
- Transaction amounts
- Account balances
- Balance changes
- Transaction frequency
- Account involvement
- Behavioural patterns

5.1 Fraud Distribution by Transaction Type

Fraud was found to be concentrated in the TRANSFER and CASH_OUT transaction types.

The approximate fraud rates identified were:

Transaction Type	Fraud Risk
TRANSFER	0.77%
CASH_OUT	0.18%

Transaction Type	Fraud Risk
PAYMENT	0% observed
CASH_IN	0% observed
DEBIT	0% observed

This finding is important because it allows the investigation and modelling process to focus more heavily on transaction types where fraudulent activity actually occurs.

However, transaction type alone is not sufficient for fraud detection.

6. Key Fraud Patterns

6.1 Full Balance Drain

One of the strongest fraud patterns identified was when:

The transaction amount is equal to the origin account's opening balance.

This means that the account is effectively emptied in a single transaction.

This behaviour is highly suspicious because it represents a complete balance drain.

A useful engineered feature for this behaviour is:

```
full_balance_drain =
amount == oldbalanceOrg
```

This feature can be used as a strong fraud indicator.

The analysis found that this pattern appeared in approximately 97.8% of fraudulent transactions while being extremely rare among legitimate transactions.

This makes it one of the strongest individual fraud indicators identified in the analysis.

6.2 Frozen Destination Balance

Another important pattern was identified when the destination account had:

```
newbalanceDest == oldbalanceDest
```

In particular, a destination account that remains at zero before and after a transaction may represent a suspicious pass-through or mule-account pattern.

This can be represented using:

```
frozen_destination =
oldbalanceDest == 0 and newbalanceDest == 0
```

This behaviour was significantly more common in fraudulent transactions than legitimate transactions.

The pattern may indicate that money is being transferred to an account that does not behave like a normal active receiving account.

7. Feature Engineering

Raw transaction variables do not always directly represent the behaviour of an account.

For this reason, additional behavioural features were created.

Examples include:

7.1 Balance Drain Ratio

```
balance_drain_ratio =  
amount / (oldbalanceOrg + 1)
```

This measures how much of the origin account's balance is being transferred.

A value close to 1 indicates that the transaction consumes almost the entire available balance.

7.2 Origin Balance Error

```
origin_balance_error =  
oldbalanceOrg - amount - newbalanceOrig
```

This measures the difference between the expected and observed origin balance.

Large inconsistencies may indicate unusual transaction behaviour or data irregularities.

7.3 Destination Balance Change

```
destination_balance_change =  
newbalanceDest - oldbalanceDest
```

This measures how much the destination balance changed after the transaction.

7.4 Origin Balance Change

```
origin_balance_change =  
oldbalanceOrg - newbalanceOrig
```

This measures the amount removed from the origin account.

7.5 Full Balance Drain Indicator

```
full_balance_drain =  
amount == oldbalanceOrg
```

This is a binary feature indicating whether the transaction completely drains the origin account.

7.6 Frozen Destination Indicator

```
frozen_destination =  
(oldbalanceDest == 0) &  
(newbalanceDest == 0)
```

This identifies destination accounts that remain at zero balance after receiving a transaction.

7.7 Transaction Type Risk

Transaction type can also be transformed into model-readable features.

For example:

```
is_transfer  
is_cash_out
```

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These features allow the model to learn that certain transaction categories have greater fraud risk.

8. Account-Level Analysis

A transaction-level model identifies suspicious individual transactions.

However, fraud often involves behaviour across multiple transactions.

For this reason, account-level analysis is also important.

The project examines account involvement by considering:

- Number of transactions
- Total transaction value
- Number of transfers
- Number of cash-outs
- Number of suspicious transactions
- Frequency of full balance drains
- Frequency of unusual balance patterns

This allows us to distinguish between:

■ *A high-volume legitimate account*

and:

An account whose activity is concentrated around suspicious behaviours.

The key insight is:

High transaction activity does not automatically imply fraud.

An account may be highly active because it is a legitimate business or financial service.

Therefore, account activity must be evaluated together with behavioural indicators.

9. Machine Learning Approach

The project uses a supervised machine learning approach.

The objective is to classify transactions as either:

0 = Legitimate
1 = Fraudulent

The general modelling pipeline is:

```
Raw Dataset
  ↓
Data Cleaning
  ↓
Feature Engineering
  ↓
Train/Test Split
  ↓
Model Training
  ↓
Fraud Prediction
  ↓
Model Evaluation
  ↓
Explainable Risk Analysis
```

The model is trained using transaction-level features and engineered behavioural indicators.

Because the fraud class is highly imbalanced, the modelling process must avoid relying on accuracy alone.

10. Model Evaluation

The model should be evaluated using metrics that are appropriate for rare-event detection.

Precision

Precision answers:

Of the transactions flagged as fraudulent, how many were actually fraudulent?

High precision means fewer false alarms.

Recall

Recall answers:

Of all actual fraudulent transactions, how many did the model detect?

High recall is important because missed fraud can result in financial losses.

F1-Score

The F1-score combines precision and recall.

It is useful when both:

- Detecting fraud
- Avoiding excessive false alarms

are important.

Precision-Recall AUC

PR-AUC is especially useful for this project because fraud is extremely rare.

It focuses on the model's ability to distinguish the minority fraud class from legitimate transactions.

11. Why Accuracy Is Not Enough

Suppose a dataset contains:

```
99.87% legitimate transactions
0.13% fraudulent transactions
```

A model that predicts every transaction as legitimate could achieve approximately 99.87% accuracy.

However:

```
Fraud detected = 0
```

Therefore, the model would be useless for fraud prevention.

This demonstrates why the project prioritises fraud-focused metrics such as:

- Recall
- Precision
- F1-score
- PR-AUC

rather than accuracy alone.

12. Explainable Fraud Detection

A major objective of ForensIQ is explainability.

A model should not only produce:

```
Fraud probability: 94%
```

It should also provide evidence such as:

```
High-risk transaction because:
```

- ✓ Transaction type: TRANSFER
- ✓ Full account balance drained
- ✓ Destination balance remained unchanged
- ✓ Transaction amount is unusually large
- ✓ Behaviour is consistent with known fraud patterns

This allows investigators to understand why a transaction was flagged.

The system therefore aims to provide:

- Evidence
- Explanation
- Risk prioritisation
- Business context

This transforms the system from a simple prediction model into a transaction intelligence tool.

13. Business Recommendations

Based on the analysis, financial institutions should consider:

13.1 Prioritising TRANSFER and CASH_OUT Transactions

These transaction types showed the strongest association with fraud in the dataset.

They should therefore receive additional monitoring.

13.2 Monitoring Full Balance Drains

Transactions that completely empty an account should receive a high-risk score.

This is particularly important when combined with other suspicious behaviours.

13.3 Monitoring Destination Accounts

Accounts that receive funds but show unusual or unchanged balance behaviour should be investigated.

13.4 Using Risk-Based Investigation

Not every alert should receive the same priority.

A better system should rank transactions based on risk:

```
graph TD;
    A[High Risk] --> B[Immediate investigation];
    C[Medium Risk] --> D[Additional verification];
    E[Low Risk] --> F[Normal monitoring];
```

This helps investigators focus their limited time on the most suspicious activity.

14. Recommended Fraud Detection Strategy

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The proposed ForensIQ system combines machine learning with interpretable rules.

The system can be represented as:

```
graph TD;
    A[Transaction] --> B[Feature Engineering];
    B --> C[Machine Learning Model];
    C --> D[Fraud Probability];
    D --> E[Behavioural Rules];
    E --> F[Risk Score];
    F --> G[Explanation];
    G --> H[Investigator Action];
```

For example:

```
graph TD;
    A[Transaction] --> B[ ];
```

```
TRANSFER
↓
Full balance drained
↓
Destination balance unchanged
↓
High fraud probability
↓
HIGH RISK
↓
Investigate immediately
```

This approach combines the strengths of:

Machine Learning

Good at identifying complex patterns.

Rules

Good at explaining known suspicious behaviour.

Human Investigation

Required for final decisions and contextual judgement.

15. Limitations

The dataset is synthetic and therefore may not perfectly represent real-world financial crime.

The following limitations should be considered:

- The data is simulated.
- Fraud patterns may be simplified.
- The dataset may not represent all real-world fraud behaviours.
- The observed absence of fraud in some transaction types does not prove that those transaction types can never be fraudulent in reality.
- Model performance may change when applied to new datasets.
- Machine learning predictions should support, not replace, human investigation.

ForensIQ should therefore be considered a prototype for transaction intelligence rather than a production-ready banking system.

16. Future Improvements

Future versions of the system could include:

Real-Time Fraud Monitoring

Transactions could be scored immediately as they occur.

Network Analysis

Accounts could be represented as nodes in a transaction network.

This could help identify:

- Mule networks
- Repeated transfers
- Circular money movement
- Coordinated fraud activity

Explainable AI

Tools such as SHAP could be used to show the features that contributed most strongly to individual predictions.

Investigator Dashboard

A dashboard could display:

- Total transactions
- Fraud rate
- High-risk transactions
- Fraud by transaction type
- Risk trends over time
- Individual transaction explanations

Continuous Monitoring

The system could monitor changes in fraud patterns over time.

This would help identify new forms of suspicious behaviour.

17. Conclusion

The ForensIQ project demonstrates how data science can be used to improve financial crime detection.

The analysis shows that fraud is not simply determined by the type of transaction or the amount of money involved.

Instead, suspicious activity is better understood through combinations of behavioural signals.

Important indicators include:

- Transaction type
- Full balance drains
- Unusual origin account behaviour
- Frozen destination balances

- Transaction amounts
- Account-level activity patterns

The central lesson from the project is:

Fraud detection should not only identify suspicious transactions. It should explain why they are suspicious and help investigators decide what to do next.

ForensIQ therefore aims to transform transaction monitoring from a simple alert-generation process into an explainable, risk-based financial intelligence system.

The proposed solution combines:

Data
+
Feature Engineering
+
Machine Learning
+
Explainability
+
Business Intelligence

to create a more practical approach to financial crime detection.

Ultimately, ForensIQ is designed not to replace investigators, but to give them faster, clearer, and more actionable intelligence.

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